RAK15005 WisBlock 128kByte FRAM Module Datasheet

Overview

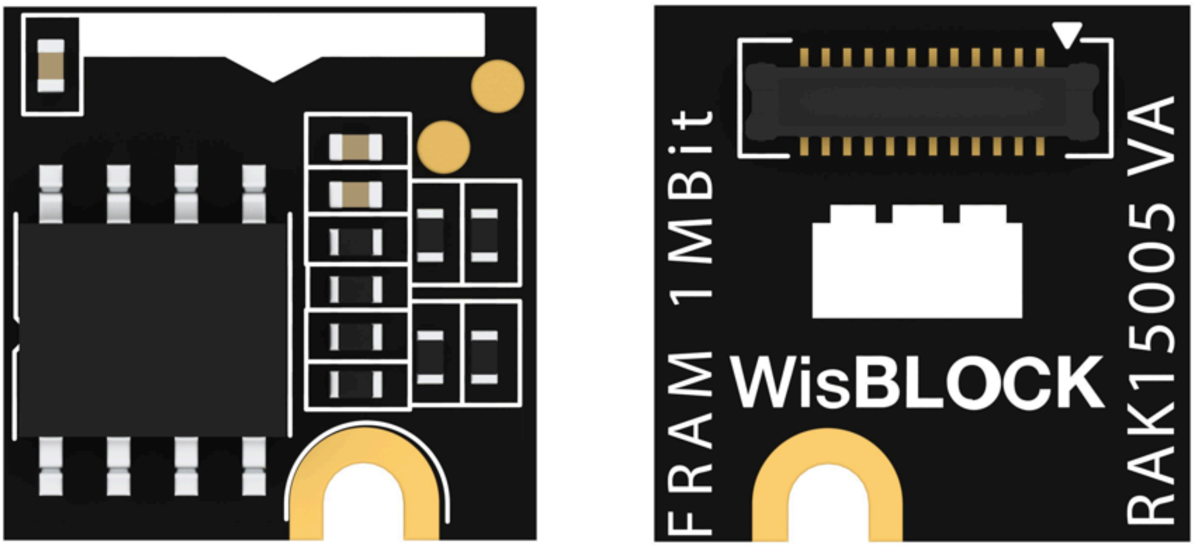

Figure 1: RAK15005 WisBlock 128kByte FRAM Module

Description

RAK15005 is a WisBlock 128kByte FRAM module that extends the WisBlock system with a MB85RC1MTPNF memory module from Fujitsu. This module is interfaced via I2C. Additionally, it can be mounted to the sensor slot of the WisBlock Base board.

Features

- Sensor specifications
  - Temperature Range: -40 °C to +85 °C
  - I2C compatible digital interface, supports 3.4 MHz
  - Operating power supply current:

    - 0.71 mA (Typ @ 3.4 MHz)
    - 1.2 mA (Max @ 3.4 MHz)
  - Standby current 15 uA (Typical)
  - 131,072 words x 8 bits
  - High Reliability：
    - Read/write endurance：10,000,000,000,000/byte
    - Data retention：
      - 10 years(+85 °C)
      - 95 years(+55 °C)
      - Over 200 years（+35 °C）

- **Size**

  - 10 × 10 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK15005 module on a WisBlock Base board. The RAK15005 module can be mounted on the slots: **A, C, D, E, & F**.

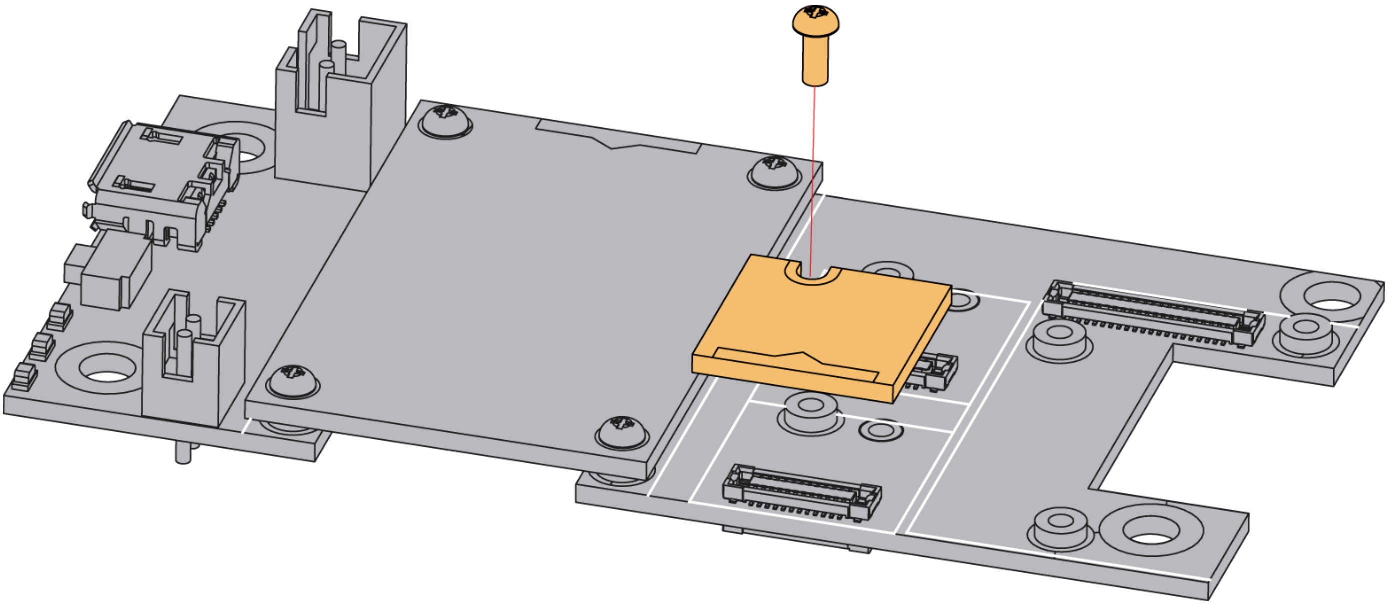

Figure 2: RAK15005 WisBlock 128kByte FRAM Module Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK15005 WisBlock 128kByte FRAM Module.

Chipset

Vendor	Part Number
Fujitsu	MB85RC1MTPNF

Pin Definition

The RAK15005 WisBlock 128kByte FRAM Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK15005 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition are shown in **Figure 3**.

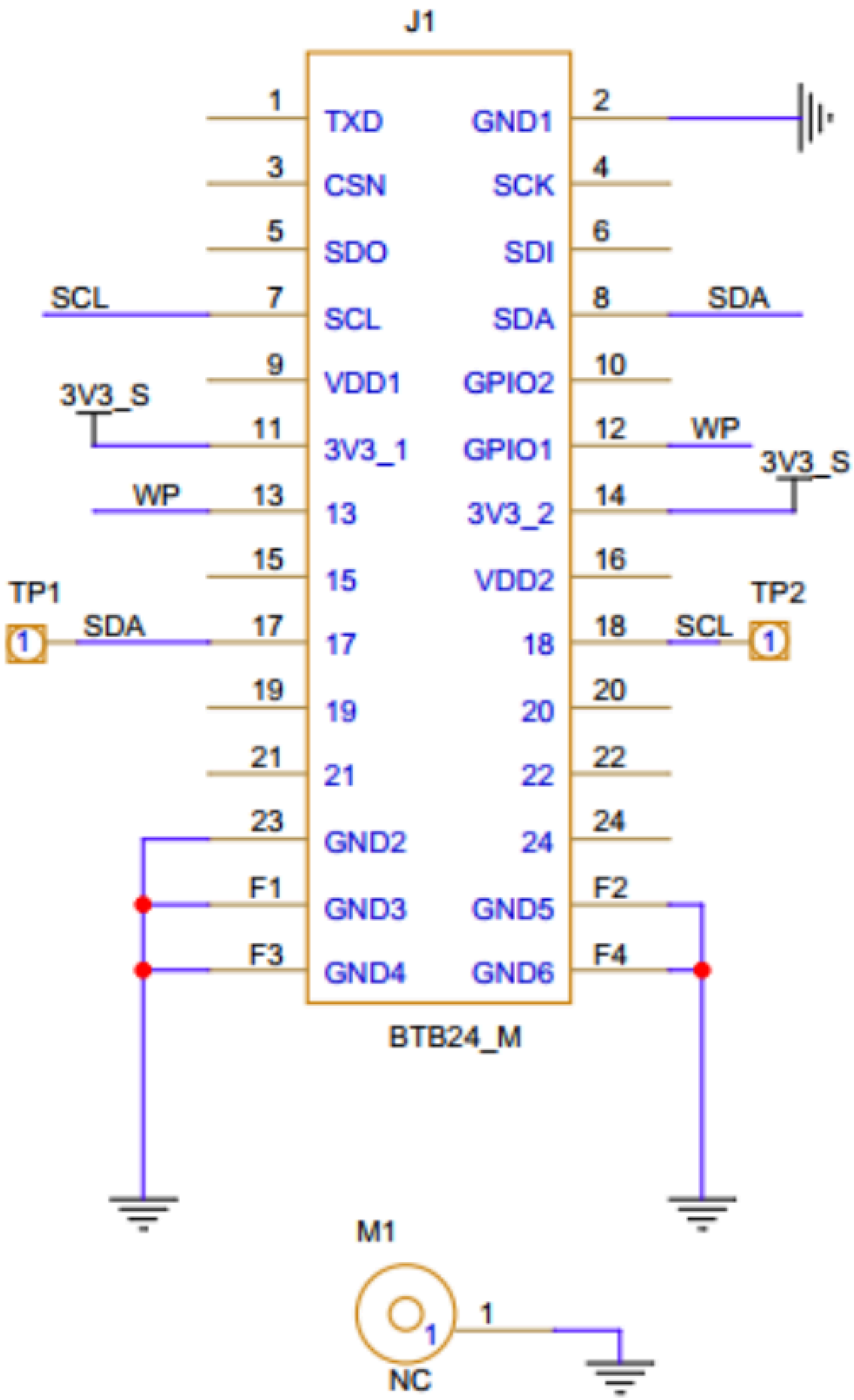

Figure 3: RAK15005 WisBlock 128kByte FRAM Module Pinout Diagram

The **WisBlock Sensor** connector is also compatible for this module. IO used for **WP** pin at **Pin 12** will depend on what sensor slot the module is plugged in. The table below shows the compatible pins used by different sensor slots:

WP (Write Protect Pin)

Slot A	Slot C	Slot D	Slot E	Slot F
IO1	IO3	IO5	IO4	IO6

> **NOTE**
>
> **RAK15005** should not be plugged in **Slot B** as it is dedicated for **IO2 (WisBlock IO2 pin)** that controls the **3V3_S** voltage output.

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VDD	Power supply for the module		3.3		V
Istb	Standby current	-	-	15	uA
Icc	Supply current, Read	-	-	710	uA
Icc	Supply current, Write	-	-	710	uA

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK15005 module.

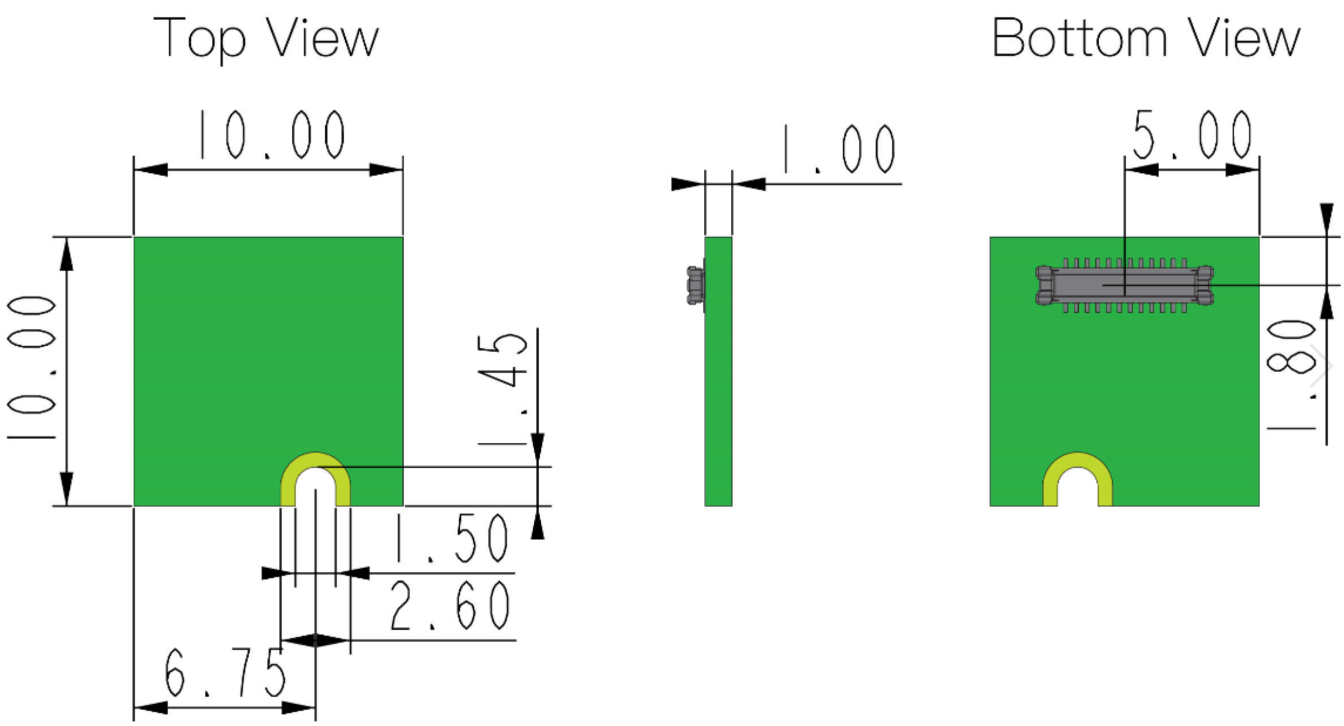

Figure 4: RAK15005 WisBlock 128kByte FRAM Module Mechanic Drawing

WisConnector PCB Layout

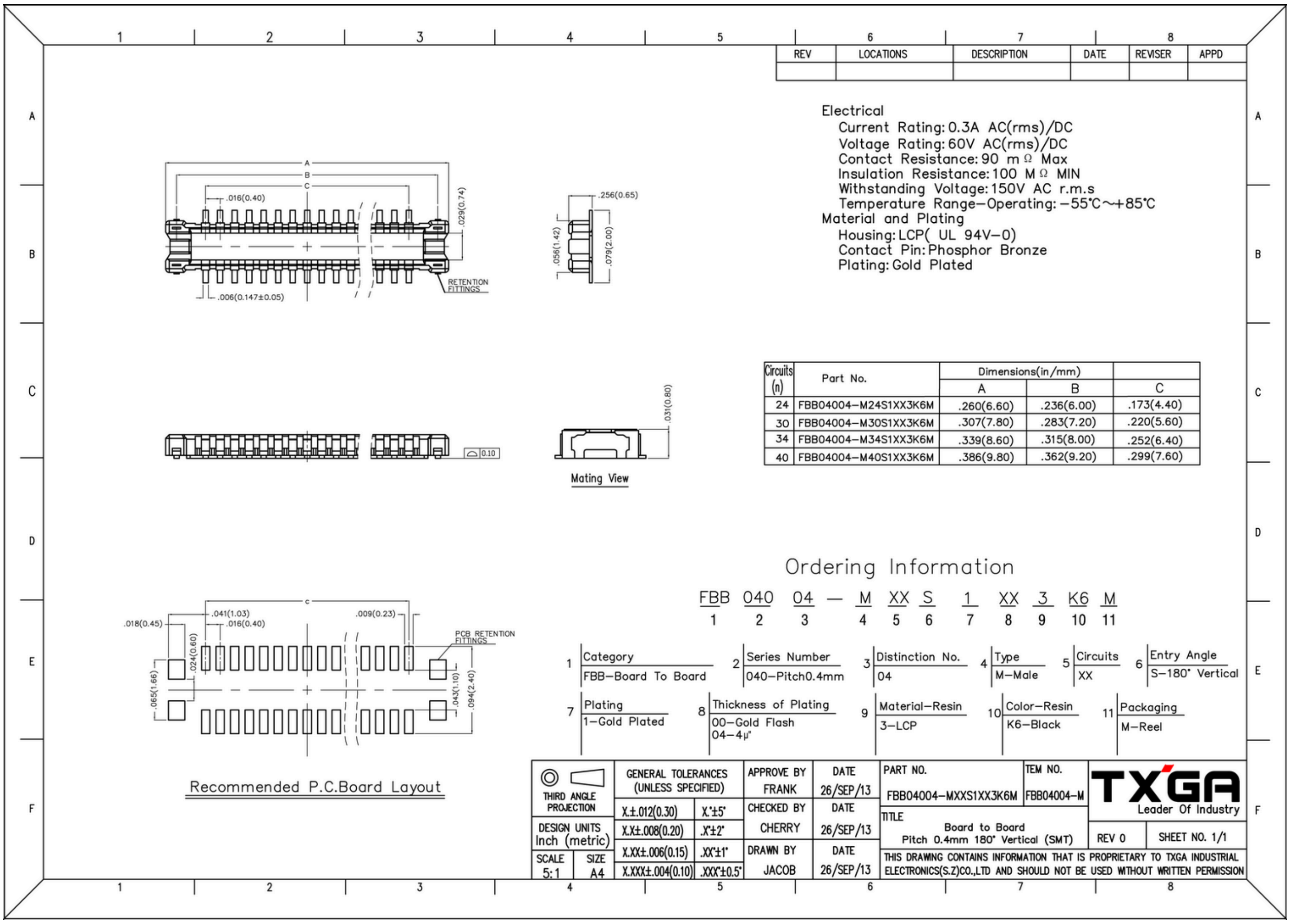

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK15005 module.

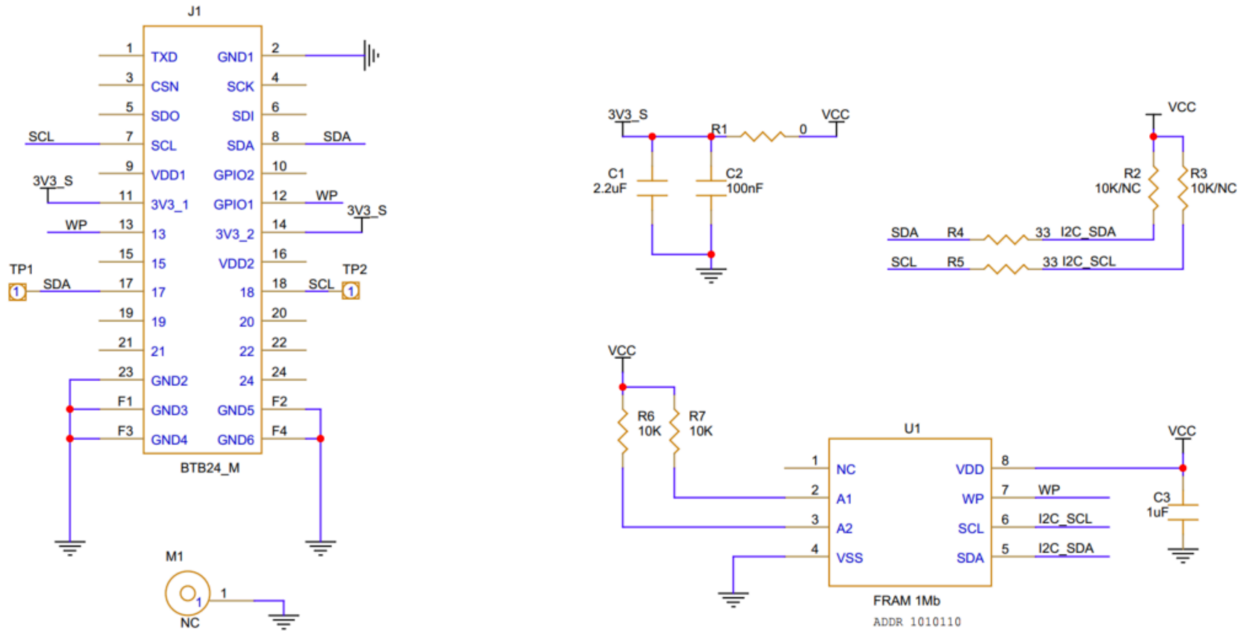

Figure 6: RAK15005 WisBlock 128kByte FRAM Module schematics